

Poisson Distribution

ANSWER KEY



Conditions for a Poisson distribution

- 1 State the key conditions required for a random variable to be modeled by a Poisson distribution.
Events occur independently of one another; events occur at a constant average rate over the given interval of time or space; two events cannot occur at exactly the same instant; the number of events in non-overlapping intervals is independent.
- 2 Explain why the number of texts a person receives in a randomly chosen hour might reasonably be modeled by a Poisson distribution, but the number of texts received right around a pre-arranged deadline might not.
Texts received during a typical hour occur independently and at a roughly constant average rate, satisfying the Poisson conditions. Texts clustered around a pre-arranged deadline (e.g. everyone replying just before a cutoff) violate the constant-rate and independence assumptions, since there is a systematic surge rather than random, uniform occurrence.

Computing Poisson probabilities

- 3 A shop receives an average of 3 customers per minute, modeled by $X \sim \text{Po}(3)$. Find $P(X = 5)$, to three decimal places.
$$P(X = 5) = \frac{e^{-3} 3^5}{5!} = \frac{(0.049787)(243)}{120} \approx 0.101.$$
- 4 Using the same shop, find $P(X = 1)$, to three decimal places.
$$P(X = 1) = 3e^{-3} \approx 3(0.049787) \approx 0.149.$$
- 5 Using the same shop, find $P(X \leq 2)$.
$$P(X = 0) = e^{-3} \approx 0.0498. P(X = 1) \approx 0.1494. P(X = 2) = \frac{9e^{-3}}{2} \approx 0.2240. \text{ Sum: } P(X \leq 2) \approx 0.423.$$

Mean and variance of a Poisson distribution

- 6 State the mean and variance of $X \sim \text{Po}(\lambda)$, and find the standard deviation when $\lambda = 9$.
 $E(X) = \lambda$ and $\text{Var}(X) = \lambda$ — both equal λ , unlike most other distributions where mean and variance are independent parameters. When $\lambda = 9$: standard deviation $= \sqrt{9} = 3$.
- 7 A random variable $X \sim \text{Po}(7)$. State $E(X)$, $\text{Var}(X)$, and the standard deviation, to three decimal places.
 $E(X) = 7$, $\text{Var}(X) = 7$, standard deviation $= \sqrt{7} \approx 2.646$.

Sums of independent Poisson variables

- 8 Two independent Poisson processes have rates $\lambda_1 = 4$ and $\lambda_2 = 6$ per hour. State the distribution of the total number of events, and find its mean and variance.
The sum of independent Poisson variables is itself Poisson: total $\sim \text{Po}(4 + 6) = \text{Po}(10)$. Mean = 10; variance = 10.
- 9 Three independent Poisson processes have rates $\lambda_1 = 2$, $\lambda_2 = 4$, $\lambda_3 = 5$ per hour. Find the mean and variance of the total number of events across all three processes in one hour.
Total $\sim \text{Po}(2 + 4 + 5) = \text{Po}(11)$. Mean = 11; variance = 11.

Poisson as an approximation to the binomial

- 10 Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the value of λ to use.
The approximation is appropriate when n is large and p is small (a common guideline: $n > 50$ and $np < 5$), so events are rare relative to the number of trials. Then X is approximately $\text{Po}(\lambda)$ with $\lambda = np$.
- 11 A factory produces 2000 items, each independently defective with probability 0.0025. Using the Poisson approximation, find the probability that exactly 2 items are defective, to three decimal places.
 $\lambda = np = 2000(0.0025) = 5$. $P(X = 2) = \frac{e^{-5} 5^2}{2!} = \frac{(0.006738)(25)}{2} \approx 0.084$.
- 12 A survey mails 5000 questionnaires, each independently containing a printing error with probability 0.0006. Using the Poisson approximation, find the probability that no questionnaires contain an error, to four decimal places.
 $\lambda = np = 5000(0.0006) = 3$. $P(X = 0) = e^{-3} \approx 0.0498$.

Using cumulative Poisson probabilities

- 13 A call center receives calls at $\lambda = 6$ per 10-minute interval. Find the probability of receiving more than 3 calls, given $P(X = 0) \approx 0.00248$, $P(X = 1) \approx 0.01487$, $P(X = 2) \approx 0.04462$, $P(X = 3) \approx 0.08924$.
 $P(X > 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)] = 1 - 0.15121 \approx 0.849$.
- 14 A website receives visitors at $\lambda = 4$ per minute. Find $P(1 \leq X \leq 3)$, given $P(X = 0) \approx 0.0183$, $P(X = 1) \approx 0.0733$, $P(X = 2) \approx 0.1465$, $P(X = 3) \approx 0.1954$.
 $P(1 \leq X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3) = 0.0733 + 0.1465 + 0.1954 \approx 0.415$.

Finding an unknown parameter

- 15 A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.0498$. Find λ , to the nearest whole number.
 $e^{-\lambda} = 0.0498 \Rightarrow -\lambda = \ln(0.0498) \approx -3.00 \Rightarrow \lambda \approx 3$.

- 16 A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 2. Find λ , and then find $P(X = \lambda)$, to three decimal places.

Since variance $= \lambda$ and standard deviation $= \sqrt{\lambda} = 2$, $\lambda = 4$.

$$P(X = 4) = \frac{e^{-4} 4^4}{4!} = \frac{(0.018316)(256)}{24} \approx 0.195.$$

- 17 A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 1) = 0.1494$, where λ is a whole number less than 5. Using $P(X = 1) = \lambda e^{-\lambda}$, determine λ by trial.

Testing $\lambda = 3$: $3e^{-3} = 3(0.049787) \approx 0.1494$ ✓, matching the given probability. So $\lambda = 3$.

Poisson processes over different intervals and conditional probability

- 18 Machine faults occur at an average rate of $\lambda = 2$ per day. Find the probability that exactly 2 faults occur today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.1353$ and $P(X = 2) \approx 0.2707$.

$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.1353 = 0.8647$. Conditional probability:

$$P(X = 2 \mid X \geq 1) = \frac{0.2707}{0.8647} \approx 0.313.$$

- 19 A radioactive source emits particles at a rate of $\lambda = 1.2$ per second, modeled by a Poisson distribution. Find $P(X = 2)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

For a 3-second interval, $\lambda = 1.2 \times 3 = 3.6$.
$$P(X = 2) = \frac{e^{-3.6} (3.6)^2}{2!} = \frac{(0.027324)(12.96)}{2} \approx 0.177.$$

A well-designed word problem: quality control at a bottling plant

- 20 This question has four parts. Faulty caps occur at random in bottling cases with an average rate of $\lambda = 1.5$ per case. (a) Find $P(X = 0)$ and $P(X = 1)$ for a randomly selected case, to four decimal places. (b) Find the probability that a case has at least 2 faulty caps. (c) The plant inspects 40 cases per day. Using the fact that a sum of independent Poisson variables is itself Poisson, state the distribution of the total number of faulty caps found across all 40 cases in a day, and find its mean. (d) Since this total has a large λ , it can be approximated by $N(\lambda, \lambda)$. Using a continuity correction and $\Phi(1.36) \approx 0.9131$, estimate the probability that more than 70 faulty caps are found in a day.

(a) $P(X = 0) = e^{-1.5} \approx 0.2231$. $P(X = 1) = 1.5e^{-1.5} \approx 0.3347$. (b)

$P(X \geq 2) = 1 - 0.2231 - 0.3347 \approx 0.4422$. (c) Total $\sim \text{Po}(40 \times 1.5) = \text{Po}(60)$, with mean 60. (d)

Using $N(60, 60)$ with $\sigma = \sqrt{60} \approx 7.746$, and a continuity correction for "more than 70" (i.e. ≥ 71):

$$z = \frac{70.5 - 60}{7.746} \approx 1.36. P(\text{total} > 70) \approx 1 - \Phi(1.36) \approx 1 - 0.9131 \approx 0.087.$$